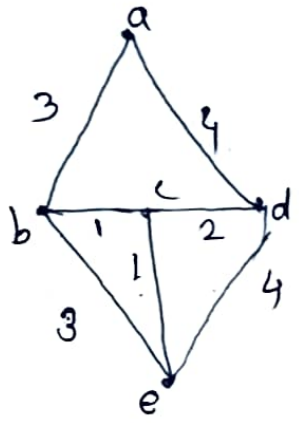
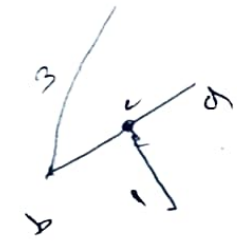
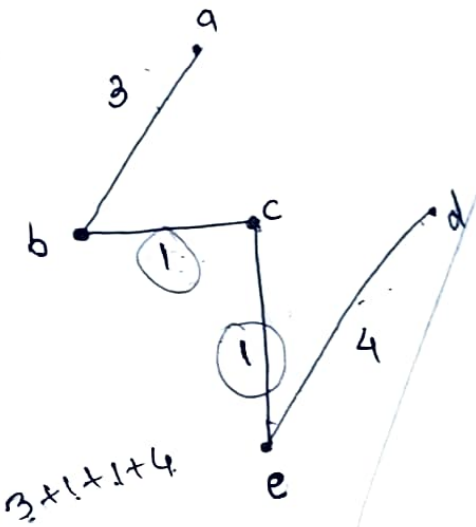
Unit - 5.

(5)

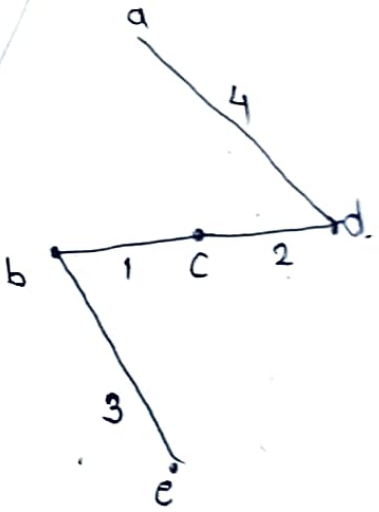
b) Apply Kruskal's algorithm to construct a minimal spanning tree for the weighted graph given below.

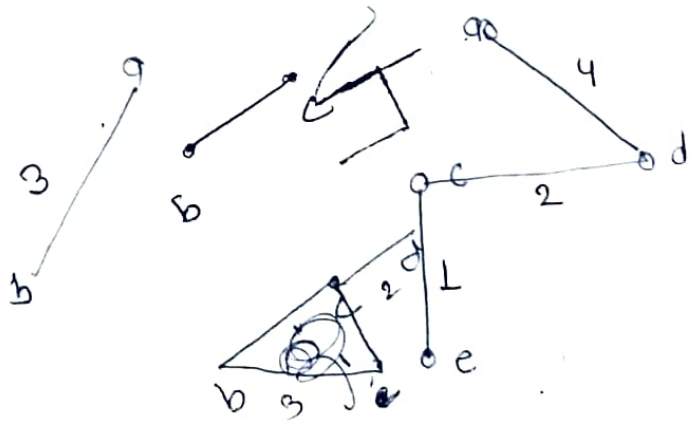


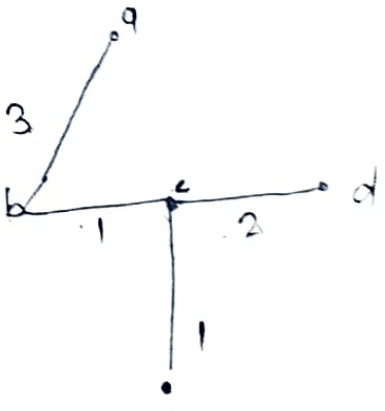




3+1+1+4

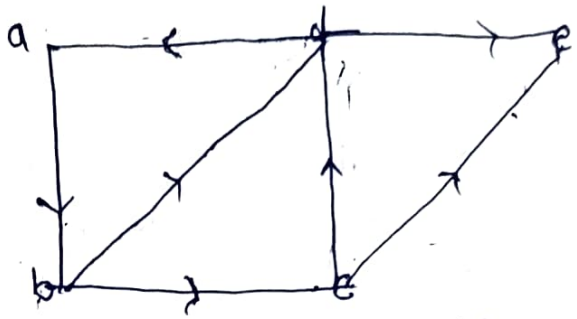






3 + 1 + 2 + 1 = 7

5] Find all indegrees and outdgrees of the graph given below



Give all elementary cycles of this graph. Is there any source or sink?

V	Indegrees	outdgres.
a	1	1
b	2	1
c	1	2
d	2	2
e	2	0

V	Indegrees	outdgrees.
a	1	1
b	1	2
c	1	2
d	2	2
e	2	0

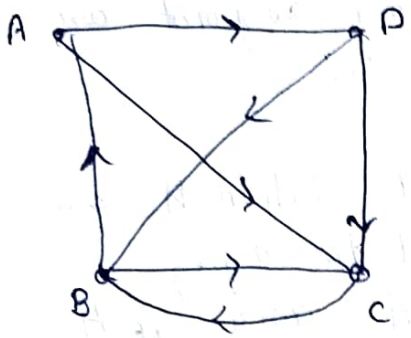
a] Draw the digraph corresponding to adjency matrix A^T & B^T

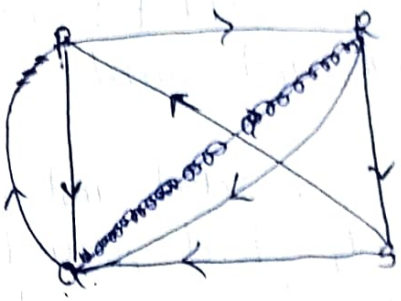
$$A = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

$$B = \begin{array}{c} \\ P \\ Q \\ R \\ S \end{array}\begin{array}{c} \begin{array}{cccc} P & Q & R & S \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

isomorphic

$$\rightarrow A^T \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \qquad B^T \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$





V	Indgree	outdgree
A	2	2
B	2	2
C	3	1
D	1	1

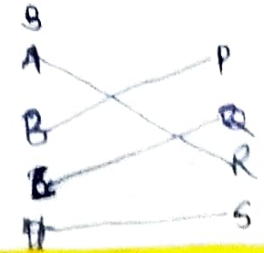
V	Indgree	outdgree
P	2	2
Q	2	2
R	2	2
S	2	2

$A = \{(1,3)\ (1,4)\ (2,1)\ (2,3)\ (3,2)\ (4,2)\ (4,3)$
$\quad\ A\ C\quad A\ D\quad B\ A\quad B\ C\quad C\ B\quad D\ B\quad D\ C$

$B = \{(1,2)\ (1,3)\ (2,1)\ (3,2)\ (3,4)\ (4,1)\ (4,2)$
$\quad\ A\ C\quad B\ A\quad C\ B$

Here $V_A = \{1, 2, 3, 4\}$

$V_B = \{1, 2, 3, 4\}$

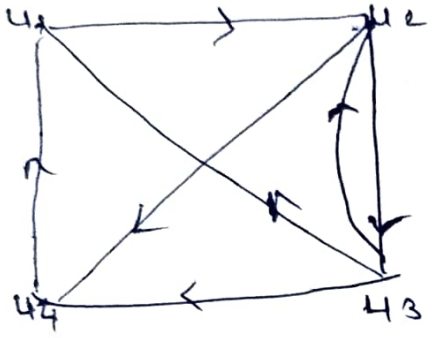


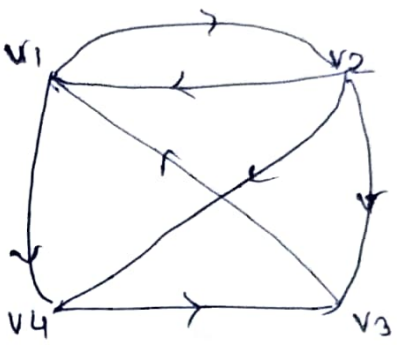
A^T =

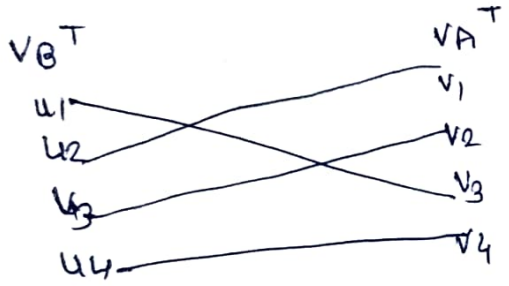
	u_1	u_2	u_3	u_4
u_1	0	1	0	0
u_2	0	0	1	1
u_3	1	1	0	1
u_4	1	0	0	0

B^T =

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

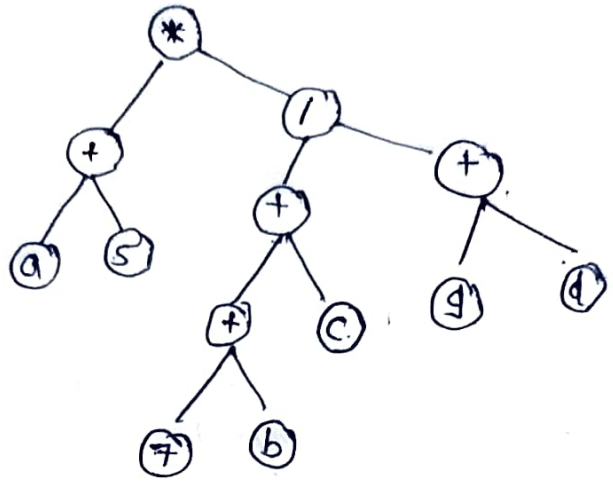




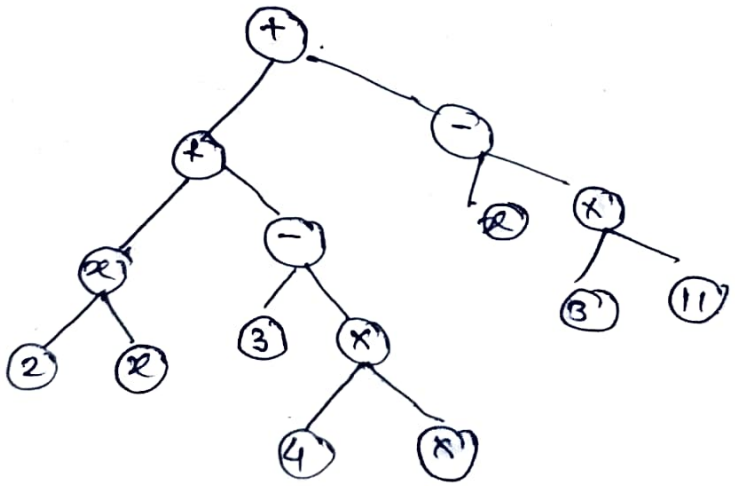


b] Draw the tree for the algebraic expression

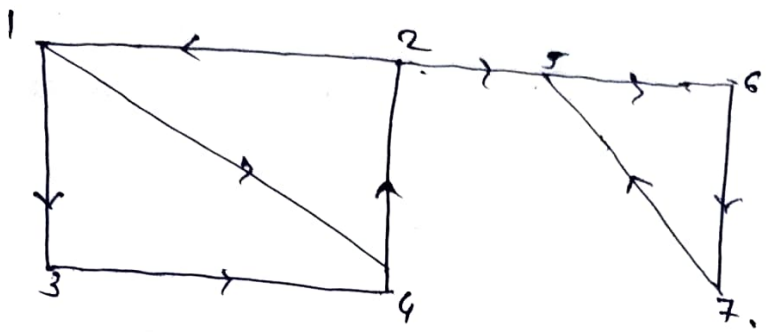
i] $(a+5) * [(7+b)+c) / (g+d)]$



ii) $(2x + (3 - 4x)) + (x - 3 \times 11))$



c) Find the node base of the following digraph. (7)



→ Here

$R(1) = \{1, 2, 3, 4, 5, 6, 7\}$
$(2) = \{2, 5, 6, 7\}$
$(3) = \{2, 3, 4, 5, 6, 7\}$
$(4) = \{2, 4, 5, 6, 7\}$
$(5) = \{5, 6, 7\}$
$(6) = \{5, 6, 7\}$
$(7) = \{5, 6, 7\}$.

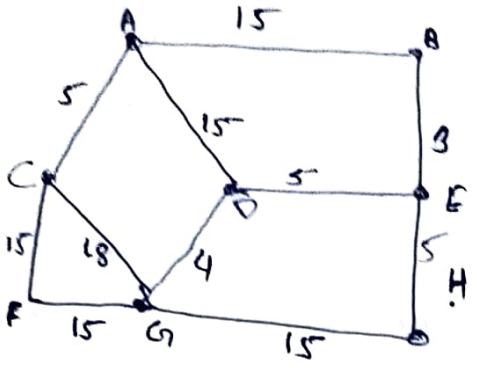
Vertex set $V = \{1, 2, 3, 4, 5, 6, 7\}$

We observe that $R(1) = V$

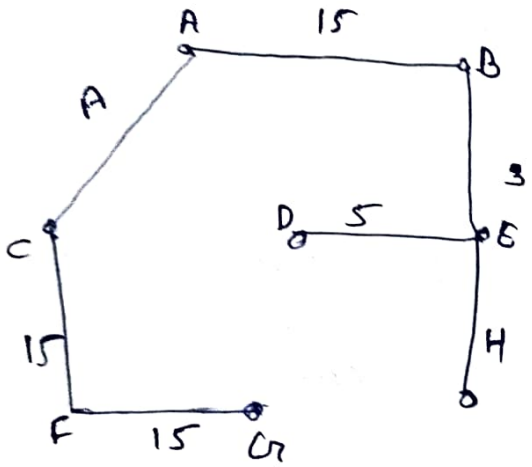
∴ Node base $= \{1\}$.

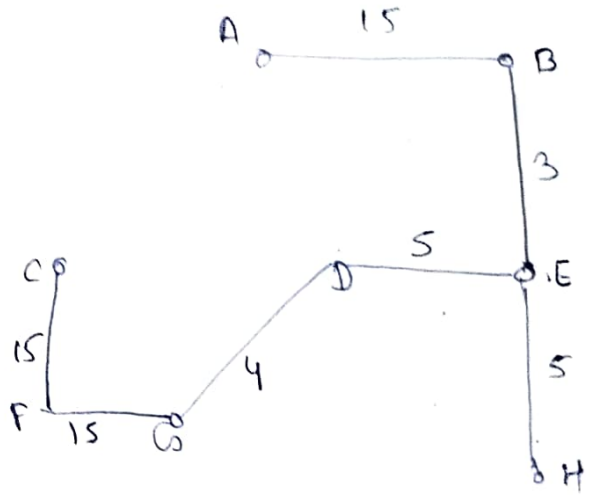
b) determine a railways network of minimal cost for the following cities using Prim's algorithm.

(8)



→





Let us select vertex B as edge (B, E) has minimum weight



Now there is one edge incident with B (B, A) and two edges incident with E (E, D) (E, H)

$W(B, A) = 15$, $W(E, D) = 5$

$W(E, H) = 5$



3) Now there is one edge incident with B (B, A) and one edge incident with E

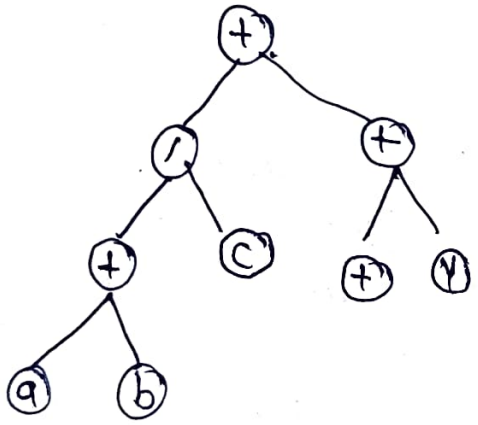


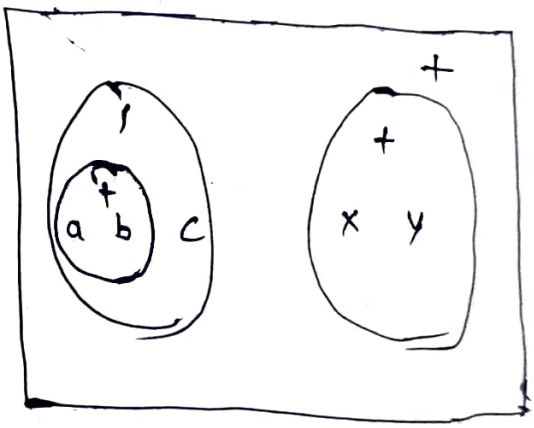
Q) Represent the following algebraic expressions using binary tree. Also draw the venn diagrams.

i) ((a+b)/c) + (x+y)

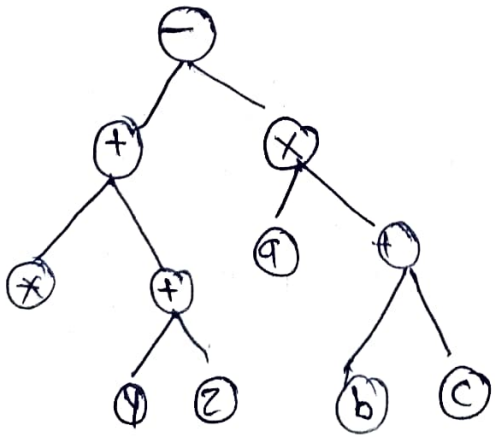
→

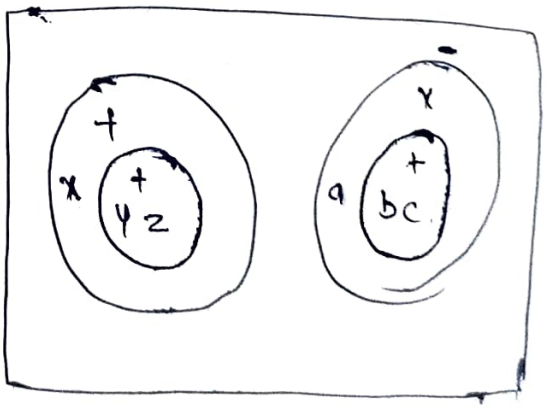
Given algebraic expression is ((a+b)/c)+(x+y) Here + is the central operator and hence it is the root.
The required binary tree and venn diagram are drawn below.



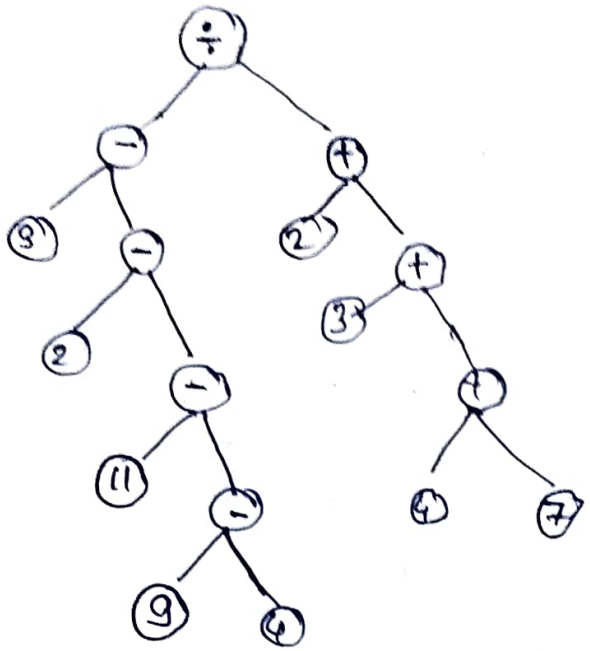


ii) ~~(x-(y+z))~~ (x+(y+z))* (ax(b+c))





iii) $(3 - 2(-11 - (9 - 4))) \div (2 + (3 + (4 + 7)))$

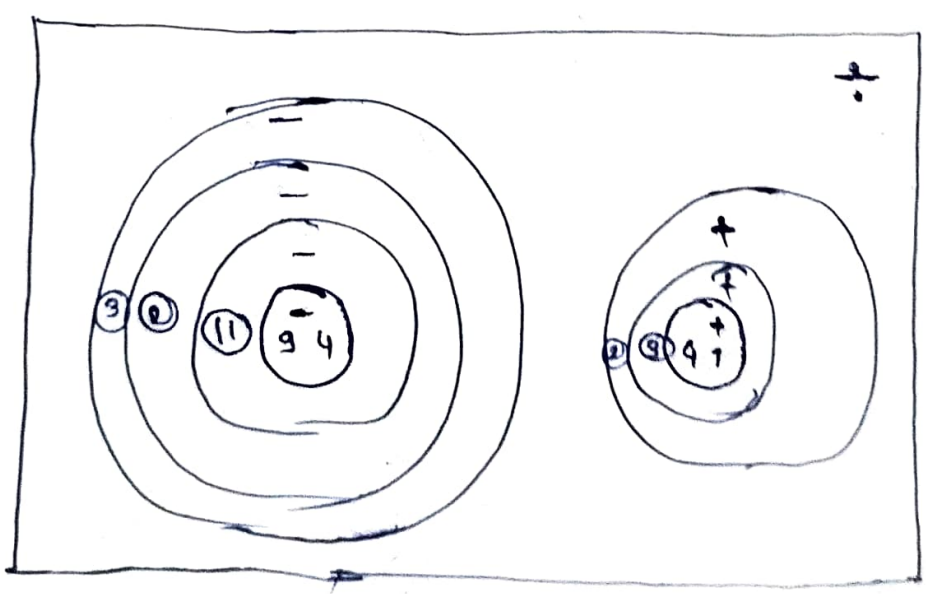


$(3 - 2(-(11 - (9 - 4)))) \div (2 + (3 + (4 + 7)))$

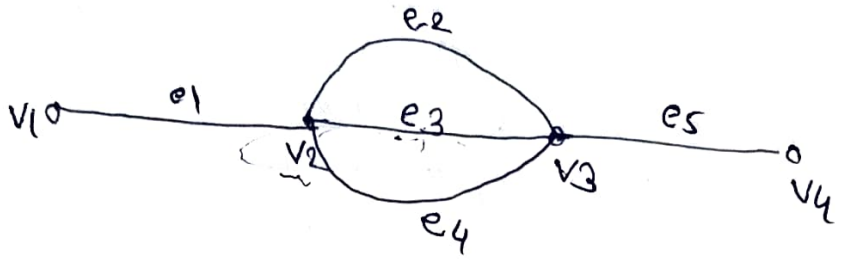
$= (3 - 2(-11 + 5))) \div (2 + (3 + 11))$

$(3 - 2 - 6) \div (2 + 14$

$= \frac{-5}{16}$



b) consider the following graph



i) How many simple paths are there from v_1 to v_4?

→ simple Path from v_1 to v_4

(v_1 e1 v_2 e_2 v_3 e_5 v_4),
(v_1 e1 v_2 e_3 v_3 e_5 v_4),
(v_1 e1 v_2 e_4 v_3 e_5 v_4). Hence there are three simple Path from v_1 to v_4

ii) How many trails are there from v_1 to v_4?

→ Trails from v_1 to v_4 are (v_1 e1 v_2 e_2 v_3 e_5 v_4),

(v_1 e1 v_2 e_3 v_3 e_5 v_4),
(v_1 e1 v_2 e_4 v_3 e_5 v_4),

(v_1 e1 v_2 e_2 v_3 e_3 v_2 e_4 v_3 e_5 v_4).
(v_1 e1 v_2 e_2 v_3 e_4 v_2 e_3 v_3 e_5 v_4).
(v_1 e1 v_2 e_3 v_3 e_2 v_2 e_4 v_3 e_5 v_4).
(v_1 e1 v_2 e_3 v_3 e_4 v_2 e_2 v_3 e_5 v_4).
(v_1 e1 v_2 e_4 v_3 e_3 v_2 e_2 v_3 e_5 v_4).
(v_1 e1 v_2 e_4 v_3 e_2 v_2 e_3 v_3 e_5 v_4).

Hence there are nine trails from v_1 to v_4.

(iii) How many Path are there from v_1 to v_4 ?

→

Path from v_1 to v_4 will be infinite, because in a path a vertex as well as an edge may repeat some of the Paths are

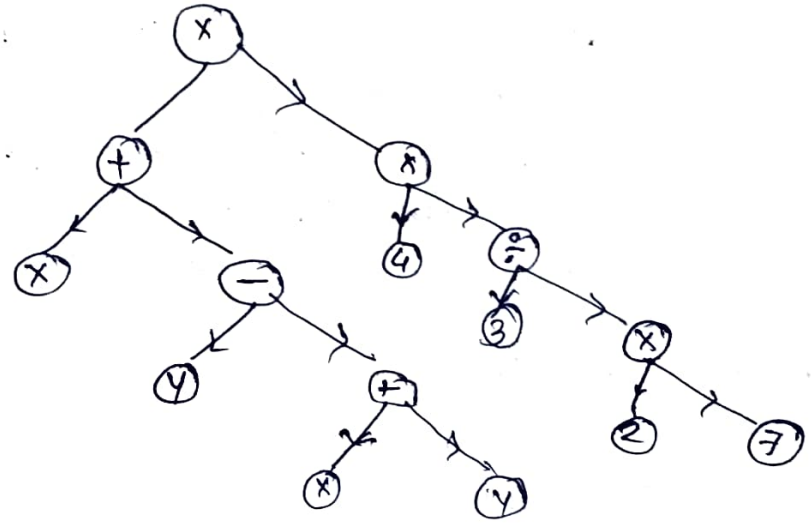
$(v_1\ e_1\ v_2\ e_2\ v_3\ e_5\ v_4)$

$(v_1\ e_1\ v_2\ e_4\ v_3\ e_3\ v_2\ e_1\ v_1\ v_2\ e_2\ v_3\ e_5\ v_4)$ etc.

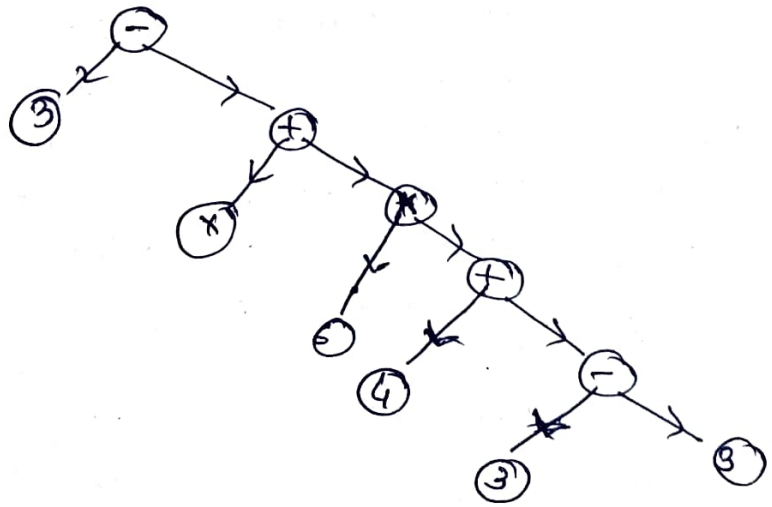
Connctendness in undirected graphs.

b) Draw binary tree for the following expression

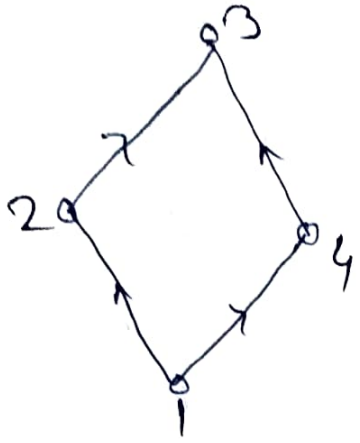
(x+(y-(x+y))) * (3÷(2x7)x4)



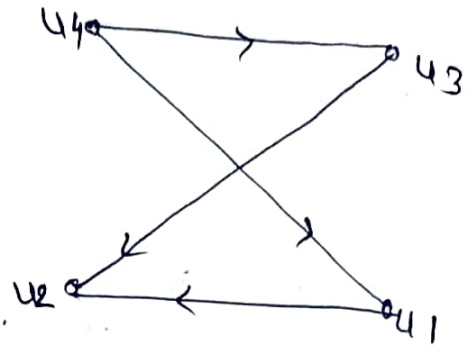
@3-(x+(6x(4-(2-3))))



c) show that the two graphs G_1 and G_2 given below are isomorphic



G_1



G_2

→

We define the correspondence as:
The matrices corresponding to the given digraphs are.

$$A = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

$$B = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Also $V_1 = V(G_1) = \{1, 2, 3, 4\}$

$V(G_2) = \{u_1, u_2, u_3, u_4\}$

$E_1 = E(G_1) = \{(1,2)\ (1,4)\ (2,3)\ (4,3)\}$

$E_2 = E(G_2) = \{(u_1,u_2)\ (u_3,u_2)\ (u_4,u_1)\ (u_4,u_3)\}$

Let $f: V_1 \longrightarrow V_2$

$f(1) = u_4$
$f(2) = u_1$
$f(3) = u_2$
$f(4) = u_3$

